K DURGA PRASAD

Software Engineer | AI/ML

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Summary

Computer Science graduate (AI/ML specialization) with a strong foundation in Data Structures & Algorithms, OOPs, DBMS, and Operating Systems. Builds RESTful APIs, backend systems, and applied AI/ML pipelines in Python, with hands-on exposure to full-stack and real-time systems. Collaborative in Agile teams with Git-based workflows and code reviews.

Education

JSS Academy of Technical Education Karnataka 2023 – Present

B.E., Computer Science & Engineering (AI/ML) | CGPA: 8.25/10 *Bengaluru, India*

Technical Skills

Languages	Python, Java, C, TypeScript, SQL, HTML/CSS
Frameworks & Libraries	Flask, FastAPI, React, REST APIs, WebSockets
AI/ML Tools	Scikit-learn, Transformers, LangChain, NLP, OpenCV, RAG (FAISS)
Tools & Platforms	Git, GitHub, Linux, Postman, VS Code, Pandas, NumPy
Core CS	Data Structures & Algorithms (DSA), OOPs, DBMS, Operating Systems (OS), Computer Networks (CN)

Projects

AI Legal Document Analyzer *GitHub | Python, FastAPI, Transformers, LangChain, FAISS*

- Architected a multi-agent ML pipeline (clause splitter, classifier, compliance, risk scoring, decision agents) automating end-to-end legal document analysis from raw PDF/DOCX input to a structured risk report.
- Integrated locally-hosted transformer models (Flan-T5) and a FAISS-based RAG system for compliance checking, eliminating dependency on external LLM APIs; added multilingual support via automatic language detection.

SyncPad — Real-Time Collaborative Text Editor *GitHub | React, TypeScript, Yjs, Hocuspocus, WebSockets*

- Built a Google-Docs-style collaborative editor using CRDTs (Yjs) for conflict-free multi-user editing and a WebSocket-based sync server (Hocuspocus) for low-latency state propagation across clients.
- Designed the system to resolve concurrent multi-user edits without data loss, a core challenge in distributed collaborative applications.

GramYatra — Real-Time Public Transport Tracker *GitHub | TypeScript, Full-Stack Web*

- Developed a real-time transit tracking and journey-planning web app for small and mid-sized cities lacking existing transit infrastructure.
- Built live vehicle tracking and route-planning features to improve commute reliability, developed as part of a hackathon team project.

Experience

RICLU Private Limited Oct 2025 (1 month)

Software Engineering Intern *Remote*

- Built and maintained RESTful API endpoints and database modules in Python/Flask, enabling seamless communication between frontend and backend services for internal applications.
- Analyzed functional requirements and optimized backend SQL queries through indexing and tuning, improving query performance by 20%.
- Wrote automated test scripts for the team's continuous integration pipeline, reducing manual debugging and verification time by 15% per sprint.
- Collaborated in an Agile team through code reviews and Git-based release management, and authored technical documentation to align modules with project requirements.

Certifications

5-Day AI Agents Intensive Course (Google/Kaggle, Dec 2025) • No Code AI Agent Builder (Simplilearn, Jan 2026) • Generative AI Studio (Google Cloud, Nov 2025) • Code Unnati 5.0 – SAP & Edunet Foundation (Ongoing)